

# TASTE SENSATION

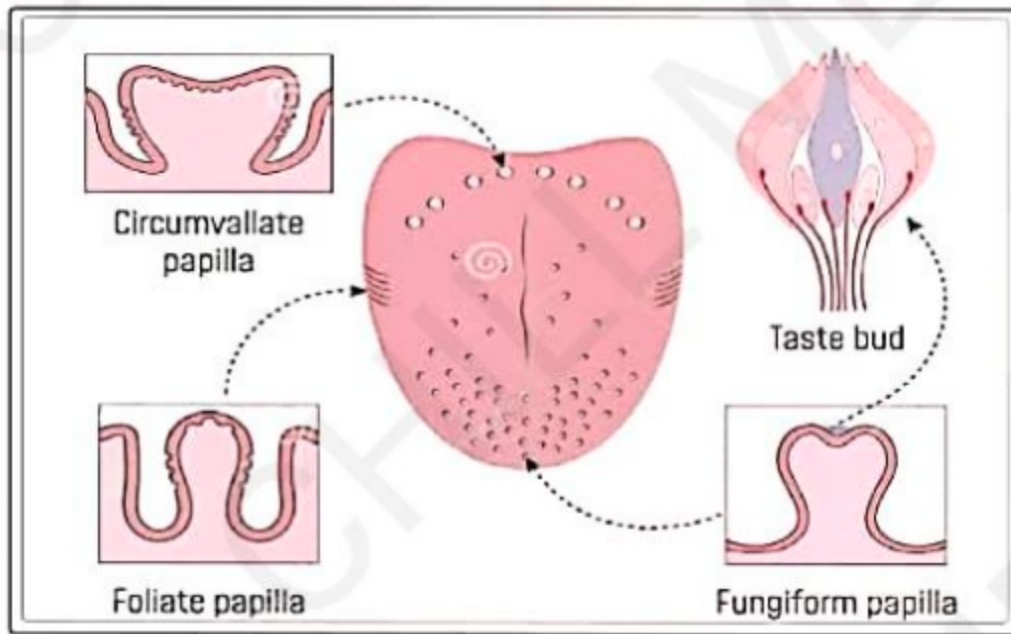
## Headings

- ▼ TASTE SENSATION
  - ▼ TASTE BUDS
    - Types of Papillae
    - STRUCTURE OF TASTE BUD
    - Types of Cells in a Taste Bud
  - ▼ PATHWAY FOR TASTE SENSATION
    - ▼ PRIMARY TASTE SENSATIONS
      1. Sweet Taste
      2. Salt Taste
      3. Sour Taste
      4. Bitter Taste
      5. Umami Taste
- APPLIED PHYSIOLOGY / CLINICAL DISORDERS

## TASTE BUDS

- **Definition** : Sense organs for taste or gustatory sensation are **taste buds**.
- **Shape** : Ovoid bodies.
- **Diameter** : 30 – 70  $\mu\text{m}$ .
- **Number** : Approximately 10,000 taste buds.
- **Distribution** : Present on the **papillae of the tongue**, and also in the mucosa of:
  - Epiglottis
  - Palate
  - Pharynx
  - Proximal oesophagus

# Types of Papillae



| Type of Papilla   | Shape & Characteristics     | Location on Tongue                                           |
|-------------------|-----------------------------|--------------------------------------------------------------|
| (a) Filiform      | Small, conical              | Over the anterior two-thirds of the tongue surface           |
| (b) Fungiform     | Round, mushroom-shaped      | Near the tip and margins of the tongue                       |
| (c) Circumvallate | Large, prominent cup-shaped | Posterior part of the tongue (in front of sulcus terminalis) |

## STRUCTURE OF TASTE BUD

- **Organization** : A bundle of taste receptor cells with supporting cells embedded together.
- **Cell Count** : Each taste bud contains approximately 40 cells.
- **Taste Pore** : There is a small apical opening called the **taste pore**.

# Types of Cells in a Taste Bud

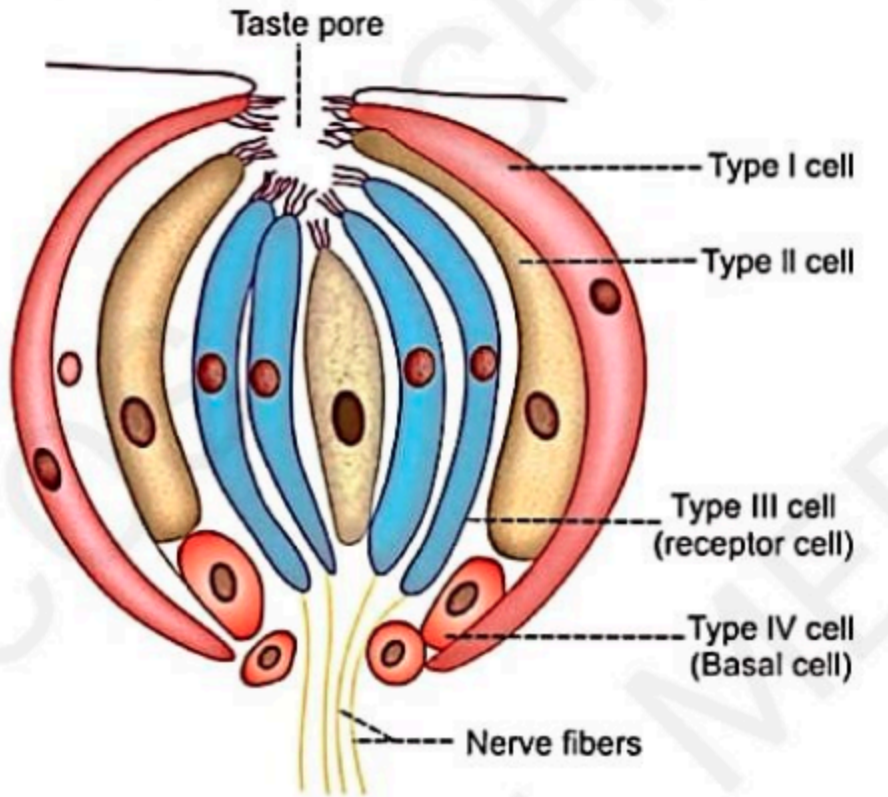
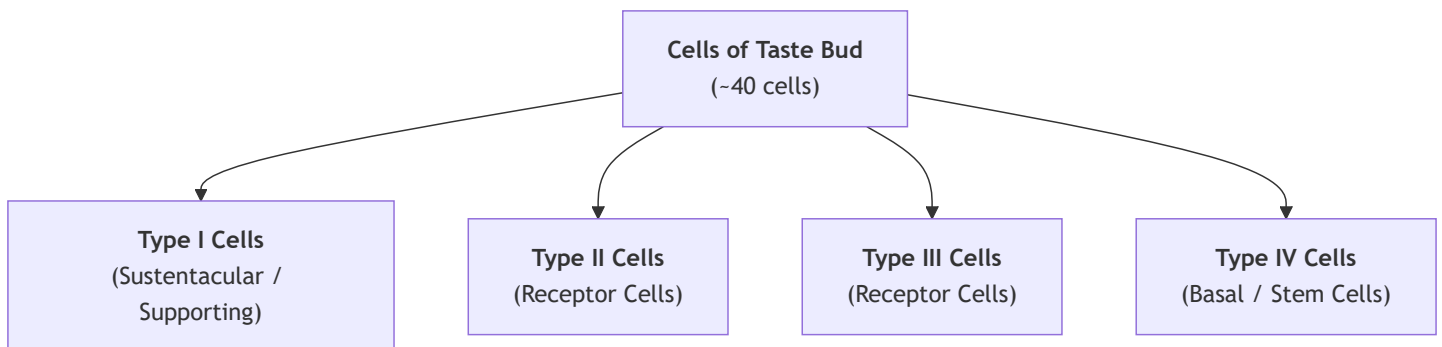
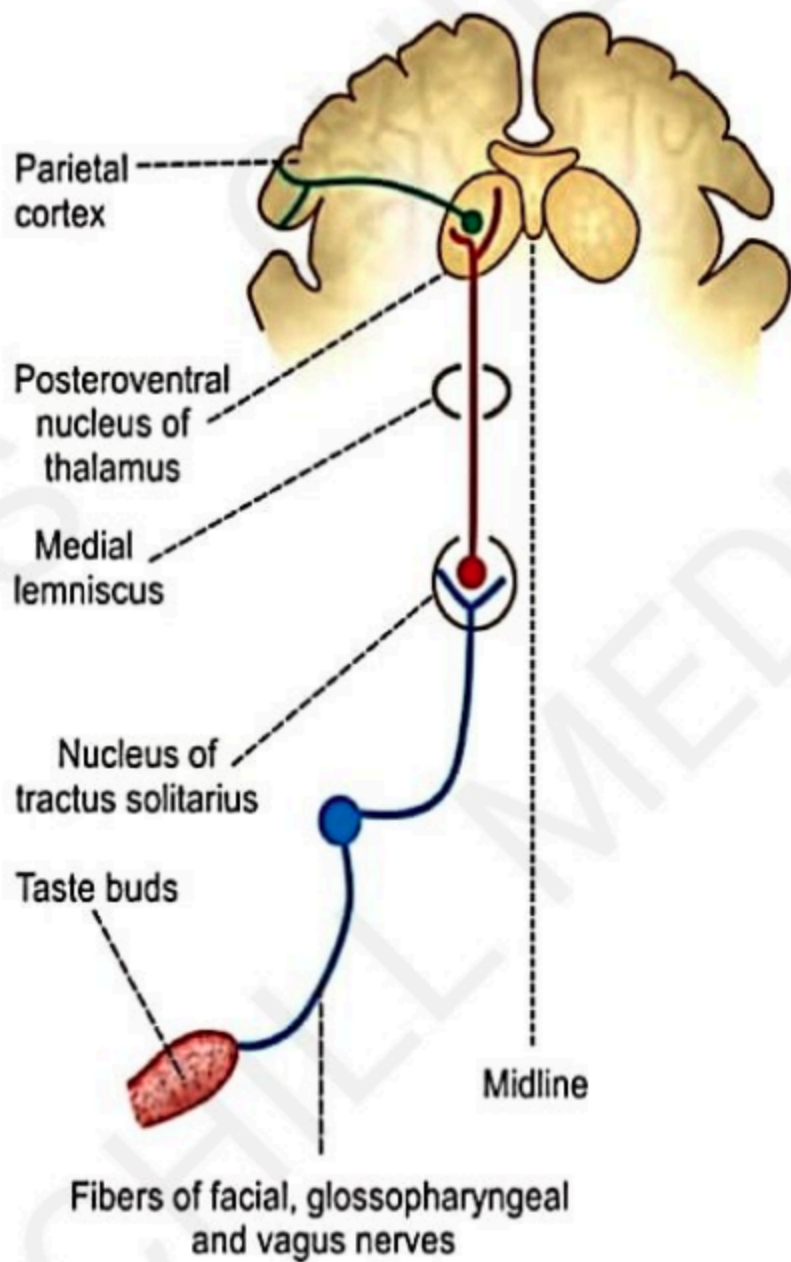


FIGURE 176.1: Taste bud



- **Functional Categorization :**
- **Receptor Cells :** Type II and Type III cells act as **taste receptor cells**.
- **Supporting Cells :** Type I and Type IV cells serve as **supporting / basal cells**.
- **Microvilli :** Type I, Type II, and Type III cells have **microvilli** projecting towards the taste pore.

# PATHWAY FOR TASTE SENSATION



**FIGURE 176.2:** Pathway for taste sensation

**Receptors (Type III Cells)**



**1st Order Neurons**

- 1) Chorda tympani fibers  
(Facial nerve - CN VII)
- 2) Glossopharyngeal nerve  
(CN IX)
- 3) Vagal fibers (CN X)



**Nucleus of Tractus  
Solitarius (NTS)**  
(Medulla)



**Medial Lemniscus**  
[2nd Order Neuron]



**Posteroventral Nucleus  
(VPM) of Thalamus**



**Parietal Lobe (Cerebral  
Cortex)**  
[3rd Order Neuron]



**Center for Taste**  
(Opercular Insular Cortex)

# PRIMARY TASTE SENSATIONS

- **5 Primary Types :**
- (a) **Sweet**
- (b) **Salt**
- (c) **Sour**
- (d) **Bitter**
- (e) **Umami**
- **Flavor Complexity :** Over **100 different tastes** can be perceived by combining these 5 primary tastes.

## 1. Sweet Taste

- **Organic Substances :** Produced mainly by mono- and polysaccharides, glycerol, alcohol, and ketones.
- **Inorganic Substances :** Lead and beryllium also impart a sweet taste.
- **Receptors & Mechanism :**
  - Receptor type is a **GPCR (G-Protein Coupled Receptor)**.
  - Sweet substances induce depolarization through the **cyclic AMP (cAMP)** pathway.

## 2. Salt Taste

- **Stimulants :** Produced by ionized salts, specifically anions and cations of **chlorides, nitrates, sulphates, bromides, and iodides**.
- **Receptor :** Epithelial Sodium Channel (ENaC).
- **Mechanism :** Influx of  $\text{Na}^+$  through ENaC causes depolarization and stimulates the release of neurotransmitters (such as glutamate).

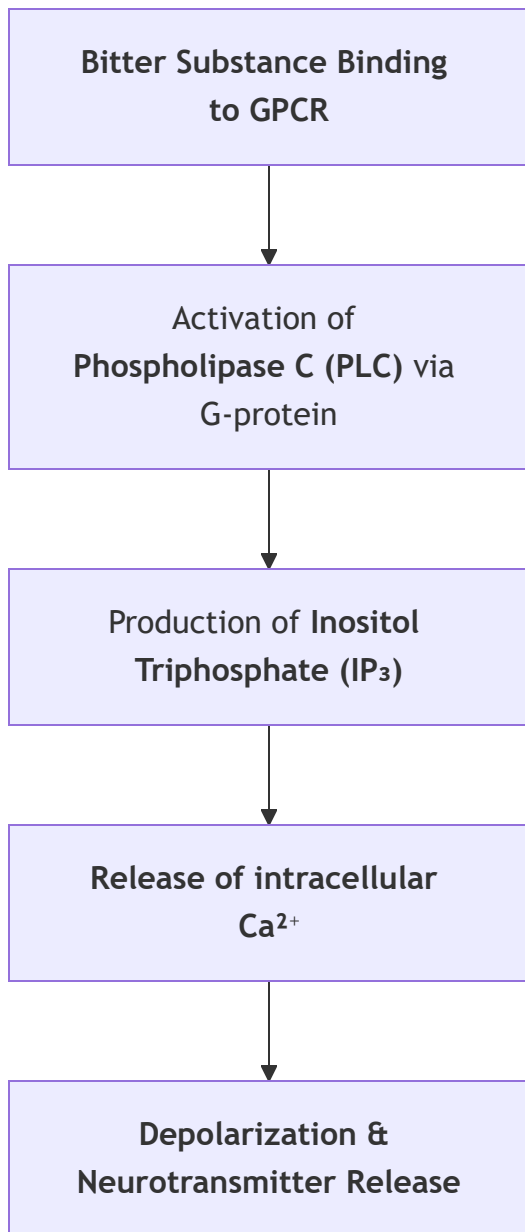
## 3. Sour Taste

- **Stimulus :** Produced by  $\text{H}^+$  ions present in acids and acid salts.
- **Receptor & Channels :**
  - Epithelial sodium channel (ENaC).

- **HCN** (Hyperpolarization-activated cyclic nucleotide-gated cation channel).
- **Mechanism** : Entry / blockage by  $H^+$  ions leads to cell depolarization.

#### 4. Bitter Taste

- **Stimulants** : Organic substances including **quinine, strychnine, morphine, picric acid, and bile salts**; certain cations also elicit a bitter sensation.
- **Receptor Type** : **GPCR** (G-Protein Coupled Receptor / **T2R** family).
- **Transduction Pathway** :



## 5. Umami Taste

- **Stimulus** : Produced when taste buds respond to glutamate, particularly **Monosodium Glutamate (MSG)**.
- **Receptor** : Metabotropic glutamate receptor (mGluR<sub>4</sub>).
- **Mechanism** : Binding of glutamate activates intracellular cascades leading to depolarization.

## APPLIED PHYSIOLOGY / CLINICAL DISORDERS

- **1. Ageusia** : Complete loss of taste sensation.
- **Lesion of Facial Nerve (CN VII) / Chorda Tympani / Mandibular Division of Trigeminal (CN V3)** : Causes loss of taste sensation in the **anterior 2/3<sup>rd</sup>** of the tongue.
- **Lesion of Glossopharyngeal Nerve (CN IX)** : Causes loss of taste sensation in the **posterior 1/3<sup>rd</sup>** of the tongue.
- **Drug-Induced (Temporary Loss)** : Caused by medications such as **captopril** and **penicillamine**.
- **2. Hypogeusia** : ↓ Decreased taste sensation.
- **3. Dysgeusia / Parageusia** : Disturbed or altered taste sensation (commonly seen in **Temporal Lobe Syndrome**).
- **4. Taste Blindness** : Rare genetic condition in which the ability to taste and recognize specific substances (e.g., Phenylthiocarbamide / PTC) is completely lost.